

Defense notes — Imagining the Target

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26 talk slide

1. Title

0:20 · cum 0:20

Good morning everyone, and thank you for being here.

2. Composed Image Retrieval

1:10 · cum 1:30

Imagine that you see a chair online that you really like, but you would like to find a dining set that includes that same chair design together with a table.

Searching only with the image of the chair would mainly return the same chair or other visually similar chairs. It would not understand that you are looking for it as part of a dining set.

On the other hand, searching only for "a dining table and chairs" would return many different sets, but probably not the specific chair you selected.

To express exactly what you are looking for, you need to combine both elements: the image identifies the chair, while the text describes the context in which you want to find it.

This is an example of composed image retrieval, which is the task I worked on in my thesis. The query consists of a reference image and a textual modification. The goal is to retrieve images that preserve the information provided by the reference image while also satisfying the modification described by the text.

3. Instance-level composed retrieval

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My thesis focuses specifically on the instance-level version of this task.

In category-level composed image retrieval, any object belonging to the correct class may satisfy the query. For example, if the query asks for a chair around a dining table, any suitable chair design might be considered correct.

In instance-level retrieval, this is not enough. The system must retrieve the exact same chair shown in the reference image, while also satisfying the textual modification.

Therefore, the main challenge is to preserve the identity of a specific object even when its context, appearance, viewpoint, or visual domain changes.

4. The i-CIR benchmark

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For our experiments, we use the i-CIR benchmark, which was specifically designed for instance-level composed image retrieval.

Each object instance in the dataset is associated with several composed queries, and every composed query consists of a reference image and a modification text.

Each query may also have multiple correct target images. This means that several database images can correctly depict the same object under the requested modification.

In addition, each instance has its own database containing carefully curated hard negatives. These are designed to confuse the retrieval system. Some may be visually similar to the reference image but fail to satisfy the textual modification. Others may satisfy the modification but depict a different object instance. Some may appear to match both parts partially while still being incorrect.

This makes the benchmark challenging and prevents the system from succeeding by relying only on the image or only on the text.

5. The i-CIR benchmark — scale

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The scale: a database of about seven hundred fifty-two thousand images, two hundred and two object instances, and one thousand eight hundred eighty-three composed queries — a median of six per instance. The histograms show the distributions.

6. Problem formulation — zero-shot

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To summarize the task, we are given a reference image and a modification text, and we want to retrieve database images that are relevant to both.

The system assigns a score to every image in the database and ranks the images from the most relevant to the least relevant.

In this thesis, we adopt a zero-shot setting. This means that we do not train or fine-tune any model on the i-CIR dataset.

This choice is important because supervised composed image retrieval requires labelled triplets consisting of a reference image, a modification text, and a correct target image. At the instance level, collecting these annotations for every new object would be expensive and difficult to scale — every newly introduced object instance could require additional labelled training data.

Instead, by using frozen pre-trained models, the retrieval system can be applied directly to previously unseen object instances without additional training.

7. Evaluation metric

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Since this is a retrieval task, the system assigns a relevance score to every image in the database. We then sort the images according to this score, from the most relevant to the least relevant. The evaluation checks where the correct target images appear in this ranked list.

For a single query, we use Average Precision, or AP. AP gives a high score when the correct images appear near the top of the ranking. If the correct images appear much lower in the list, the score decreases.

We then calculate mean Average Precision, or mAP, by averaging the AP scores across all queries.

However, the number of queries is not the same for every object instance. Some instances contribute many queries, while others contribute only a few. If we averaged directly over all queries, the instances with more queries would have a larger influence on the final result.

For this reason, our main metric is macro-mAP. We first average the query scores within each object instance, and then average across all instances. In this way, every instance contributes equally, regardless of how many queries it contains.

8. Vision-language models: CLIP and SigLIP

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The main technical challenge is combining the information from the reference image and the modification text. To represent both modalities, we use frozen, pre-trained vision-language models, specifically CLIP and SigLIP.

These models use a dual-encoder, or two-tower, architecture. One encoder processes images, while the other processes text. Both encoders map their inputs into the same embedding space. Because the image and text representations share this space, we can measure their semantic similarity using cosine similarity. An image receives a high similarity score when its

embedding is close to the embedding of the corresponding text.

The models are used completely frozen in our experiments. We only extract their embeddings; we do not update or fine-tune their parameters.

CLIP and SigLIP follow the same general dual-encoder principle, but they differ in how they are trained. CLIP uses a batch-level contrastive objective based on a softmax: for each image, the matching text must be distinguished from the other texts in the batch, and vice versa. SigLIP instead uses a pairwise sigmoid loss. Each image-text pair is treated independently as either a matching pair or a non-matching pair. This avoids the global softmax normalization used by CLIP and can scale more effectively to large training batches.

In our work, we compare these different frozen backbones to test whether the proposed method depends on one particular vision-language model.

9. The baseline: BASIC

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The baseline for our work is BASIC, the zero-shot method introduced together with the i-CIR benchmark.

BASIC processes the two parts of the composed query separately. First, the reference image is encoded by the visual encoder, and the modification text is encoded by the text encoder. The images in the database are also encoded in advance using the visual encoder.

For every database image, BASIC computes two similarity scores. The first is the visual similarity, which measures how similar the database image is to the reference image. The second is the textual similarity, which measures how well the database image matches the modification text. These similarities are computed using the dot product between the corresponding embeddings.

The two resulting scores are then multiplied. This multiplicative fusion acts like a soft logical AND: a database image should receive a high final score only if it matches both the reference image and the textual modification.

This is the core idea of BASIC. The method also includes several additional processing steps — centering, score normalization, contextualization — which I will discuss later.

10. Limitation of the baseline — our proposal

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The main limitation of the baseline is that the reference image and the modification text are never processed jointly.

The image branch only sees the reference image, while the text branch only sees the modification. As a result, the model does not explicitly represent the complete composed query.

To address this limitation, we introduce a third scoring branch based on a generated target caption.

We provide both the reference image and the modification text simultaneously to a multimodal large language model. We then ask the model to imagine that the requested modification has been applied to the object in the reference image, and to describe the resulting target image.

For example, in the chair query, the model receives the image of the specific chair together with the text "placed around a table". It then generates a caption such as: "White metallic chair with scrolled armrests placed around a table."

This caption contains information from both parts of the query: the visual identity of the chair and the context specified by the text.

We then encode the generated caption using the same frozen text encoder and compute its similarity with every image in the database. The final retrieval score therefore combines three signals: similarity to the reference image, similarity to the original modification text, and similarity to the generated target caption.

11. The captioner: InternVL 3.5-8B

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For target-caption generation, we use the frozen eight-billion-parameter version of InternVL 3.5.

We selected this model because it is an open-source multimodal large language model that can jointly process images and text.

It also provides a practical balance between performance and computational cost. Although it is much smaller than many larger multimodal models, it achieves competitive results across a range of vision-language benchmarks.

Most importantly, we use it without any fine-tuning. It only receives the reference image, the modification text, and our instruction to describe the expected target image.

12. Designing the instruction

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An important part of the caption-generation stage is the instruction given to the multimodal model.

The quality of the generated caption depends strongly on how the task is formulated. For this reason, I carried out an extensive prompt-design process and tested more than twenty different instructions.

I varied several elements, including the level of detail requested, the use of examples, the order of the constraints, and restrictions intended to prevent the model from introducing irrelevant information.

Based on these experiments, I selected five complementary prompting strategies. Each strategy encourages the model to describe the imagined target from a slightly different perspective, and together they produce a diverse set of candidate captions.

13. Effect of the caption branch

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Here, we can see that adding a single generated caption improves macro-mAP by about 5.9 points, from 17.95 to 23.83. This suggests that the caption captures useful compositional information that is not fully represented by the image and text branches separately.

Using multiple captions improves the result further. We generate five captions using different prompting strategies, average their embeddings, and use the resulting representation for retrieval. This reduces the influence of prompt-specific wording or irrelevant details and emphasizes information that is consistent across the captions.

With this five-caption average, macro-mAP reaches 25.41, an improvement of about 7.5 points over the original image-text baseline.

| *From this point on, all results use CLIP ViT-L/14 as the backbone, unless stated otherwise.*

14. The pipeline

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To summarize the pipeline so far, each composed query contains a reference image and a modification text. We also generate a caption describing the imagined target image.

These three components — the reference image, the original text, and the generated caption — are encoded using frozen, pre-trained vision-language models and mapped into a shared embedding space.

For every database image, we compute three similarity scores: its similarity to the reference image, to the modification text, and to the generated caption.

We first normalize these scores so that they are on comparable scales. We then combine them multiplicatively and apply the Harris-inspired penalty, which reduces the score of candidates that match only one or two branches strongly while performing poorly on the others.

I will now discuss two additional processing techniques that further improve the retrieval performance.

15. Combining the three scores

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Before combining the visual, textual, and caption scores, we first apply min-based normalization so that the three branches are on comparable scales. Any negative residual values are then clamped to zero, meaning they provide no positive evidence for the candidate image.

The normalized scores are combined multiplicatively, because the final result should perform well across all branches at the same time.

We also use a Harris-inspired penalty. The idea comes from the Harris corner detector, where a strong response in only one direction is not sufficient. Similarly, in our retrieval setting, a candidate should not receive a high score just because it matches one branch very well while matching the others poorly.

For example, an image may contain the exact chair but not show it around a table, or it may show chairs around a table but not the specific chair from the reference image. The Harris term penalizes this imbalance, and it is combined with a hyperparameter λ .

As a result, the final score favors candidates that receive consistently high scores from the image, text, and caption branches.

16. Post-processing I — centering

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The first technique is centering.

Vision-language embedding spaces often contain a large common component shared by many embeddings. This component may represent generic visual or linguistic patterns rather than the distinctive information required for retrieval.

To reduce this effect, we compute a mean embedding for each modality using a separate corpus. We subtract this mean from the corresponding embeddings and then normalize them again.

The average image feature reflects general visual patterns, while the average text feature captures common linguistic patterns. Subtracting them helps isolate semantic content from distributional bias.

It should not be understood as literally removing "the sky" from every image. Instead, it removes directions in the embedding space that are common across many samples, such as generic visual composition, background patterns, or broad dataset biases.

17. Post-processing II — contextualisation

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The second technique is contextualisation of the text query.

CLIP is trained primarily on natural-language captions and full sentences. As a result, using single-word text queries, such as "sculpture", or sentence fragments, such as "during sunset", constitutes out-of-distribution input and may produce text features that are not well aligned

with the image features.

To address this, we introduce a contextualisation step that enriches such textual queries with additional terms. We generate multiple caption-like queries by combining the raw text query with elements from a subject corpus, adding a random set of terms before — for example "dog during the sunset" — and after — for example "sculpture dog" — the text query.

These composed phrases are embedded using the text encoder, centered, and averaged. This yields a more robust textual feature that better reflects how the model interprets concepts in natural language.

18. Adding the steps on top of each other

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Here we add each post-processing step cumulatively, for the baseline in blue and for our caption pipeline in red.

The caption pipeline stays above the baseline along the entire ladder. Centering gives a large jump for the baseline, and contextualisation gives the largest single post-processing gain.

If asked why the ladder stops here: two further BASIC steps — the contrastive projection and query expansion — do not help on this benchmark once the caption branch is present, so they are omitted from our pipeline.

19. Vision-language backbones

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We evaluate the pipeline with four frozen backbones, spanning two model families and two capacity levels — both to measure how much the choice of backbone matters, and to report numbers comparable to the i-CIR baseline.

CLIP ViT-L/14 is the backbone most commonly reported in the zero-shot CIR literature, which makes our numbers directly comparable. CLIP ViT-H/14 is the higher-capacity variant, and the strongest CLIP model on the plain image-text baseline.

SigLIP replaces the softmax contrastive loss with the pairwise sigmoid loss. We use SigLIP ViT-L/16 at 256 pixels, and the more recent, higher-capacity SigLIP2 g/16 at 384 pixels, which is the best-aligned encoder of the four.

20. The caption branch across backbones

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The same pipeline, on all four frozen backbones.

The caption branch improves every single one of them. The largest jump is on SigLIP-Large, where macro-mAP roughly doubles, from twenty-one to forty-three. SigLIP2 is the strongest base model.

This shows that our contribution does not depend on one particular vision-language model.

21. Final results — SigLIP2

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This is the best backbone, with the full ladder. The shape is the same as on CLIP-L: the caption pipeline stays above the baseline throughout, and contextualisation gives the final jump.

Our best configuration reaches 61.98 macro-mAP, against 57.69 for BASIC at the same rung — and against 38.1 for the plain image-times-text product.

22. Our best overall system

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To state the final system plainly: the SigLIP2 backbone, the caption branch with five averaged captions, and post-processing through centering and contextualisation. Every component is frozen.

It reaches 61.98 macro-mAP and 62.07 mAP. Beyond mAP: the correct target is ranked first for 71% of queries, and it appears in the top ten for 95% of queries. Recall at one hundred is 94%.

That high recall is important — I will come back to it in the future work.

23. Qualitative example — success

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A qualitative example. The query is this Tintin drawing, with the modification "as an iconic rooftop sign".

The plain image-times-text product already reaches 89% AP, but it mixes in generic Tintin drawings that are not rooftop signs. Adding the generated caption — "Cartoon character in a beige coat and blue shirt appears as an iconic rooftop sign" — pushes the true rooftop-sign photographs to the top of the ranking: 98% AP.

24. Qualitative example — failure

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And an honest failure, on the same instance: "as a human size statue".

Here the caption branch does not help. The generated caption describes the character correctly, but the visual branch keeps pulling in drawings and figurines rather than life-size statues.

This is the characteristic failure mode: when the modification demands a drastic change of visual domain, the image branch dominates the product, and the caption is not enough to overcome it.

25. Limitations and future work

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Three limitations. The main cost is caption generation — about fifteen seconds per query, which dominates online inference. The results also depend on prompt design: averaging five captions mitigates this, but does not remove it. And the fixed, unweighted product cannot adapt when one branch should matter more than the others.

Several directions follow naturally. A stronger or more efficient captioner — or a single high-quality caption that matches the five-caption average without five generations — would cut the main compute cost. The fixed product could be replaced by a learned or weighted fusion of the three similarities, relaxing the zero-shot constraint in exchange for accuracy.

Because recall is already high — the correct instance lands in the top-ranked shortlist for the large majority of queries — a second-stage reranker that reorders this shortlist is a promising direction: most of the remaining error is in the ordering of candidates that are already retrieved. A text-conditioned reranker that reorders by similarity to the target caption, rather than to the reference image, is a natural candidate, since it keeps the modification in the loop at the reranking stage.

Finally, the caption-as-third-modality idea is benchmark-agnostic and could be evaluated beyond i-CIR — on category-level CIR benchmarks and other instance-level retrieval settings — to test how far the explicit target-description signal generalises.

26. Conclusions

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Four conclusions.

The caption is the single most effective component. Adding the target caption improves every backbone on the full dataset, and in the component ablation it is the largest single contributor — larger than feature centering, and far larger than the remaining BASIC preprocessing.

A simple caption can replace elaborate preprocessing. A plain text-times-image baseline plus

the caption reaches the level of the fully tuned BASIC pipeline, and with a strong backbone exceeds it. The textual signal is better supplied by an explicit target description than by contextualising the bare modification phrase.

Centering and the caption are complementary. Their gains add up, and the best configuration combines both.

SigLIP2 is the strongest backbone for the pipeline, reaching the thesis-best 61.98 macro-mAP with the caption, centering, and contextualisation — even though CLIP-H is stronger on the bare image-text baseline. The backbone must be judged within the full pipeline.

27. Thank you

Thank you. I am happy to take questions.

Q&A map

Full CLIP-L ablation, every rung → **backup 28**. "The paper reports 34.35 for BASIC" → **backup 29** (our reproduction gives 32.48; the gap is in the image branch, from re-extracting features in a newer software stack; both methods are evaluated in the same environment, so the comparisons hold). Latency breakdown → **backup 30**. The five instruction strategies with their captions → **backup 31**.

Metric formulas are on slide 7; the fusion details on slide 15; the backbone table on slide 19.